

Andrew Liu

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EDUCATION

University of California, Los Angeles — B.S. Computer Science + Physics, 4.0/4.0 GPA

Expected June 2028

Relevant Coursework: Data Structures, Graph Theory, Computer Organization, Honors Abstract Algebra, Beams + Accelerator Physics

Awards: USA Computing Olympiad, Gold Division (2026); USA Physics Olympiad, Silver Medal (2025); Putnam Exam, score 13 (2025)

WORK EXPERIENCE

Upscale AI — Software Engineer Intern

June 2026 – Present

- Productionized a GPU timing engine on an internal cloud EDA platform — **Go control plane**, Packer-built AMIs, OpenAPI/React frontend — with fail-closed handling of malformed tool output and content-digest provenance for reproducible worker images
- Extended the platform to timing-driven GPU place-and-route, integrating HeteroSTA and the PyTorch-based DreamPlace 4 placer through a new control-plane job type, **distributed agent executor flow**, worker script, and GPU AMI rebake
- Replaced thousands of brute-force re-simulations with a **gradient-driven cell-sizing loop** over a persistent GPU session, screening ~100 candidate swaps per pass in ~28s; validated with a 19/19 directional sign test and root-caused a regression to a stale timing baseline
- Cut end-to-end timing jobs to ~20 minutes by collapsing a **Dockerized EC2** GPU worker and a remote tool host onto one machine, with a self-validating Tcl/Python extraction pipeline running inside the live signoff session

RESEARCH EXPERIENCE

Huang Group, UCLA — Undergraduate Research Assistant

February 2026 – Present

- Built an unattended **Linux/Bash daemon** supervising 5 concurrent EPOS heavy-ion simulations that produced ~20 GB of events over multiple days with zero manual intervention — process-liveness checks before file operations, race-free harvesting into a versioned dataset, and auto-restart of finished workers
- Engineered a **C++** pipeline measuring elliptic flow across 500M+ tracks: two-pass streaming design with pre-allocated buffers for an $O(N^2)$ pair stage (~ 5×10^8 evaluations); cut wall-clock time ~10× by tuning columnar I/O to deserialize only referenced fields
- Traced a plausible but null result to a **mislabelled reaction-plane angle** by reconstructing each event's orientation, comparing candidates by correlation (0.44 vs. 0.04)

PROJECTS

Accelerator Literature Classifier — Musumeci Group, UCLA

2026

- Built a **supervised text-classification** pipeline in Python/scikit-learn that mines the Crossref API for every Physical Review Accelerators & Beams paper since 2021 and labels each as advanced vs. conventional research
- Trained **TF-IDF + logistic regression** classifiers with balanced class weights, exported the learned coefficients as an interpretable term-weight table, and tuned precision/recall against a manually corrected ground-truth set using a confusion-matrix harness

Racked — Social Discovery App for Local Thrift Stores

2026

- Built an iOS app in **React Native + TypeScript** — 20 screens spanning a geolocated discovery feed and map, social graph, 1:1/group chat, and notifications — on TanStack Query with optimistic cache updates, GitHub Actions CI, and 281 tests
- Designed a PostgreSQL schema with **row-level security** as the only authorization boundary: 49 RLS policies and 35+ security-definer RPCs enforce visibility server-side, with PostGIS distance-ranked discovery over GiST-indexed geography columns
- Found and fixed a **side-channel** where blocked users' message timing leaked through inbox ordering, replacing a trigger-maintained sort column with a block-filtered lateral join

IMC Prosperity 4 Trading Bot — Top 250 of 18,000 teams globally

2026

- Implemented EMA mean-reversion and market-making strategies in **Python**, modeled implied volatility smiles over high-volume historical data to price options, and built an interactive backtest visualizer to speed signal discovery

SKILLS

Languages: C, C++, Java, JavaScript/TypeScript, Go, Python, SQL, Tcl, Assembly

Systems & Infrastructure: Linux/Bash, AWS (EC2, AMIs), Docker, Packer, PostgreSQL/PostGIS, REST/OpenAPI, Git, GitHub Actions

Frameworks & Tools: React, React Native, scikit-learn, PyTorch, pandas, NumPy, Jest, ROOT, gdb